

Def. Let G be a Group. Define the upper central series $\{Z_n(G)\}_{n=0}^{\infty}$ as:

$$Z_0(G) = 1,$$

$$Z_1(G) = Z(G),$$

⋮

$$Z_{i+1}(G) \stackrel{\text{def}}{=} \mathcal{Q}_i^{-1}[Z(G/Z_i(G))] \quad \text{where } \mathcal{Q}_i: G \rightarrow G/Z_i(G): x \mapsto xZ_i(G)$$

$$\text{Note: } 1 = Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \dots$$

because $Z_i(G) \leq \mathcal{Q}_i^{-1}[Z(G/Z_i(G))]$ given by

$$x \in Z_i(G) \Rightarrow \mathcal{Q}_i(x) = xZ_i(G) = 1 \cdot Z_i(G) \in Z(G/Z_i(G))$$

$$\Rightarrow x \in \mathcal{Q}_i^{-1}[\mathcal{Q}_i(x)] \subseteq \mathcal{Q}_i^{-1}[Z(G/Z_i(G))]$$

and, preimage of Normal is normal, the definition is well-defined.

$$\text{Clearly, } Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G))$$

$$\begin{aligned} Z_{i+1}(G)/Z_i(G) &= \mathcal{Q}_i^{-1}[Z(G/Z_i(G))]/Z_i(G) \quad \mathcal{Q}_i \text{ onto} \\ &= \mathcal{Q}_i[\mathcal{Q}_i^{-1}[Z(G/Z_i(G))]] = Z(G/Z_i(G)) \end{aligned}$$

Def. A Group G is called Nilpotent if for some $c \geq 0$, $Z_c(G) = G$.

The smallest such $c \geq 0$ is called the nilpotent class of G .

Prop. Let P be a group of order p^a , where p is prime and $a \geq 1$.

Then P is Nilpotent of nilpotent class at most $a-1$.

Proof. Since P -group has no trivial center, for all $i \geq 0$,

$$|P/Z_i(P)| > 1 \Rightarrow Z(P/Z_i(P)) \neq 1$$

$$\text{Therefore, } Z_i(P) \neq P \Rightarrow |P/Z_i(P)| > 1 \Rightarrow Z(P/Z_i(P)) \neq 1 \Leftrightarrow Z_{i+1}(P)/Z_i(P) > 1$$

This implies $|Z_{i+1}(P)| \geq p \cdot |Z_i(P)|$, further, $|Z_{i+1}(P)| \geq p^{i+1}$

In particular, $|Z_a(P)| \geq p^a$, at least $i = a$, $Z_a(P) = P$.

If nilpotent class of P is exactly a , then for all $i \geq 0$, $|Z_i(P)| = p^i$.

Then, $P/Z_{a-2}(P)$ has order p^2 , so is abelian. Now,

$$Z_{a-1}(P) = \mathcal{Q}_{a-2}^{-1}[Z(P/Z_{a-2}(P))] = P$$

which is Contradiction.

upper, lower Central series.